

Lesson 4: The Greeks

Analyst onboarding · Lesson 4 of 10 · L/S Gamma Desk, QUANTT

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Goal: measure how an option's price reacts to SPY moving, time passing and volatility changing.

You need: Lessons 1–3 (options, implied volatility).

1 Why we need the Greeks

An option's price changes for three main reasons: SPY moves, a day passes, or implied volatility changes. The **Greeks** are named after Greek letters, and each one measures one of these sensitivities. They tell you **what risks you hold** right now.

Our running example: the **761 call, 29 DTE, SPY at 760, IV 14%**.

Greek	Measures	Our 761 call (per share)	Per contract ($\times 100$)
Delta (Δ)	Price change per \$1 SPY move	0.50	+50 shares of SPY exposure
Gamma (Γ)	Change in delta per \$1 SPY move	0.013	+1.3 shares per \$1
Theta (Θ)	Price change per day passing	−0.29	−\$29 per trading day
Vega (ν)	Price change per 1 vol point	0.87	+\$87 per vol point

2 Delta: direction

Delta is how much the option's price moves when SPY moves \$1.

- An ATM call has delta about **0.5**: if SPY rises \$1, the call gains about \$0.50 per share, or **\$50 per contract**.
- That is the same exposure as owning **50 shares** of SPY. Traders call this “50 deltas”.
- Calls have delta between 0 and 1, and puts between −1 and 0. An ATM put has delta about −0.5.
- Deep ITM options behave like the stock (delta near ± 1). Far OTM options barely react (delta near 0).

⇒ Delta is the option's **direction bet**. Our desk removes it by trading shares (Lesson 5).

3 Gamma: how fast delta changes

Gamma is how much the **delta itself** changes when SPY moves \$1.

- Our call has gamma 0.013. If SPY rises \$10, delta goes from 0.50 to about 0.63. The option now behaves like 63 shares instead of 50.
- **Gamma is largest for ATM options** and for options **close to expiry**, and shrinks for far ITM or OTM strikes.
- **Buying options gives positive gamma** (you are “long gamma”). **Selling gives negative gamma** (“short gamma”).

⇒ Long gamma means your exposure **grows in the direction SPY moves**: you get longer as it rises and shorter as it falls. This is the curved payoff from Lesson 2, and it is the thing our desk is named after.

4 Theta: the cost of time

Theta is how much the option loses as one day passes, everything else unchanged.

- Our call loses about \$0.29 per share, or **\$29 per contract per trading day**.
- Theta speeds up as expiry approaches: the last week loses value fastest.
- **Buyers pay theta. Sellers collect it.**

5 Vega: sensitivity to implied volatility

Vega is how much the option's price changes when IV moves one vol point.

- Our call has vega 0.87: if IV rises from 14% to 15%, the call gains about **\$87 per contract**, even if SPY doesn't move.
- **Longer-dated options have more vega.** Short-dated options have more gamma.

6 The key relationship: gamma and theta are two sides of one coin

You can't get gamma for free. For a hedged position, theta and gamma are linked by:

$$\Theta \approx -\frac{1}{2} \Gamma S^2 \sigma_{IV}^2$$

Read it as: **the market sets the option's price so that the theta you pay exactly covers the gamma you get, if SPY moves exactly as much as IV says.** Whether you win or lose then comes down to one question: does SPY move **more** or **less** than implied? Lesson 5 makes this exact.

7 Calls vs puts once delta is removed

Put-call parity links a call and a put with the same strike and expiry:

$$C - P = S - Ke^{-rT}$$

A call minus a put equals SPY minus some cash. So a call and a put differ only by a straight stock position, which has **no gamma, theta or vega**. After you hedge away the delta, a call and a put at the same strike carry **almost the same gamma, theta and vega**. The real differences are practical: skew (Lesson 3), liquidity, and early exercise.

⇒ This is why our desk can use a **call** for long gamma and a **put** for short gamma (Lesson 8). Once hedged, both are “the same” volatility bet.

8 Greeks of a whole position

Greeks add up. Your position's delta is the sum of every option's delta $\times 100 \times$ number of contracts, plus any shares you hold (each share has delta 1).

Example: you hold 1 long 761 call (+50 deltas) and are short 50 SPY shares (−50). Your net delta is **0**: small SPY moves no longer change your P&L. But your gamma is still +1.3 shares per \$1, so the next move will make delta non-zero again.

9 In our code

- IBKR sends Greeks with each option quote. `pipeline/chain.py` stores delta, gamma, vega and theta per contract.
- `core/Algos/Hedging/portfolio.py` (`option_delta`) adds up the delta of every SPY option we hold.

10 Words to know

- **Delta:** direction exposure, in shares. **Gamma:** how fast delta changes.
- **Theta:** daily time decay. **Vega:** sensitivity to IV.
- **Long / short gamma:** own options / have sold options. **Put-call parity:** links calls and puts.

11 Check yourself

1. You sold 1 ATM put (delta -0.5). What is your position delta, in shares?
2. Which has more gamma: a 7-DTE ATM option or a 60-DTE ATM option?
3. IV jumps 3 points overnight while SPY is flat. Roughly what does our 761 call gain?

Answers: (1) +50: selling a -0.5 delta put $\times 100$ gives +50. (2) The 7-DTE option. (3) About $3 \times \$87 = \261 .